

Package Insert

1. Name of the product

Remedica Co-Amoxiclav 625 mg film-coated tablets

2. Name and Strength of Active Ingredient

Each film-coated tablet contains amoxicillin trihydrate equivalent to 500 mg amoxicillin and diluted potassium clavulanate equivalent to 125 mg clavulanic acid.

3. Dosage form

Film-coated tablet

White, oval, film-coated tablets.

4. Pharmacological Properties

Pharmacodynamics

Pharmacotherapeutic Group: Antibacterials for systemic use; Beta-lactam antibacterials, penicillins, ATC Code: J01CR02

Mechanism of action

Amoxicillin is a semisynthetic penicillin (beta-lactam antibiotic) that inhibits one or more enzymes (often referred to as penicillin-binding proteins, PBPs) in the biosynthetic pathway of bacterial peptidoglycan, which is an integral structural component of the bacterial cell wall. Inhibition of peptidoglycan synthesis leads to weakening of the cell wall, which is usually followed by cell lysis and death.

Amoxicillin is susceptible to degradation by beta-lactamases produced by resistant bacteria and therefore the spectrum of activity of amoxicillin alone does not include organisms which produce these enzymes.

Clavulanic acid is a beta-lactam structurally related to penicillins. It inactivates some beta-lactamase enzymes thereby preventing inactivation of amoxicillin. Clavulanic acid alone does not exert a clinically useful antibacterial effect.

Pharmacokinetic/pharmacodynamics relationship

The time above the minimum inhibitory concentration ($T > MIC$) is considered to be the major determinant of efficacy for amoxicillin.

Mechanisms of resistance

The two main mechanisms of resistance to amoxicillin/clavulanic acid are:

- inactivation by those bacterial beta-lactamases that are not themselves inhibited by clavulanic acid, including class B, C and D.
- alteration of PBPs, which reduce the affinity of the antibacterial agent for the target.

Impermeability of bacteria or efflux pump mechanisms may cause or contribute to bacterial resistance, particularly in Gram-negative bacteria.

Breakpoints

MIC breakpoints for amoxicillin/clavulanic acid are those of the European Committee on Antimicrobial Susceptibility Testing (EUCAST).

Organism	Susceptibility Breakpoints (µg/ml)		
	Susceptible	Intermediate	Resistant
<i>Haemophilus influenza</i> ¹	≤ 1	-	> 1
<i>Moraxella catarrhalis</i> ¹	≤ 1	-	> 1
<i>Staphylococcus aureus</i> ²	≤ 2	-	> 2
Coagulase-negative staphylococci ²	≤ 0.25		> 0.25
<i>Enterococcus</i> ¹	≤ 4	8	> 8
<i>Streptococcus A, B, C, G</i> ⁵	≤ 0.25	-	> 0.25
<i>Streptococcus pneumoniae</i> ³	≤ 0.5	1-2	> 2
Enterobacteriaceae ^{1,4}	-	-	> 8
Gram-negative Anaerobes ¹	≤ 4	8	> 8
Gram-positive Anaerobes ¹	≤ 4	8	> 8
Non-species related breakpoints ¹	≤ 2	4-8	> 8

¹ The reported values are for Amoxicillin concentrations. For susceptibility testing purposes, the concentration of Clavulanic acid is fixed at 2 mg/l.

² The reported values are Oxacillin concentrations.

³ Breakpoint values in the table are based on Ampicillin breakpoints.

⁴ The resistant breakpoint of R>8 mg/l ensures that all isolates with resistance mechanisms are reported resistant.

⁵ Breakpoint values in the table are based on Benzylpenicillin breakpoints.

The prevalence of resistance may vary geographically and with time for selected species, and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Commonly susceptible species

<u>Aerobic Gram-positive micro-organisms</u> <i>Enterococcus faecalis</i> . <i>Gardnerella vaginalis</i> . <i>Staphylococcus aureus</i> (methicillin-susceptible) £ Coagulase-negative staphylococci (methicillin-susceptible). <i>Streptococcus agalactiae</i> . <i>Streptococcus pneumoniae</i> ¹ . <i>Streptococcus pyogenes</i> and other beta-haemolytic streptococci <i>Streptococcus viridans</i> group
<u>Aerobic Gram-negative micro-organisms</u> <i>Capnocytophaga</i> spp. <i>Eikenella corrodens</i> . <i>Haemophilus influenzae</i> . ² . <i>Moraxella catarrhalis</i> . <i>Pasteurella multocida</i> .
<u>Anaerobic micro-organisms</u> <i>Bacteroides fragilis</i> . <i>Fusobacterium nucleatum</i> . <i>Prevotella</i> spp.
Species for which acquired resistance may be a problem
<u>Aerobic Gram-positive micro-organisms</u> <i>Enterococcus faecium</i> \$.
<u>Aerobic Gram-negative micro-organisms</u> <i>Escherichia coli</i> . <i>Klebsiella oxytoca</i> . <i>Klebsiella pneumoniae</i> . <i>Proteus mirabilis</i> . <i>Proteus vulgaris</i> .
Inherently resistant organisms
<u>Aerobic Gram-negative micro-organisms</u> <i>Acinetobacter</i> sp. <i>Citrobacter freundii</i> . <i>Enterobacter</i> sp. <i>Legionella pneumophila</i> . <i>Morganella morganii</i> . <i>Providencia</i> spp. <i>Pseudomonas</i> sp. <i>Serratia</i> sp. <i>Stenotrophomonas maltophilia</i> .
<u>Other micro-organisms</u> <i>Chlamydophila pneumoniae</i> . <i>Chlamydophila psittaci</i> . <i>Coxiella burnetti</i> .

Mycoplasma pneumoniae.

§ Natural intermediate susceptibility in the absence of acquired mechanism of resistance.

£ All methicillin-resistant staphylococci are resistant to amoxicillin/clavulanic acid

¹ *Streptococcus pneumoniae* that are resistant to penicillin should not be treated with this presentation of amoxicillin/clavulanic acid (see sections Posology and method of administration and Warnings and Precautions).

² Strains with decreased susceptibility have been reported in some countries in the EU with a frequency higher than 10%.

Pharmacokinetics

Absorption

Amoxicillin and clavulanic acid, are fully dissociated in aqueous solution at physiological pH. Both components are rapidly and well absorbed by the oral route of administration. Absorption of amoxicillin/clavulanic acid is optimized when taken at the start of the meal. Following oral administration, amoxicillin and clavulanic acid are approximately 70% bioavailable. The plasma profiles of both components are similar and the time to peak plasma concentration (T_{max}) in each case is approximately one hour.

Distribution

About 25% of total plasma clavulanic acid and 18% of total plasma amoxicillin is bound to protein. The apparent volume of distribution is around 0.3-0.4 l/kg for amoxicillin and around 0.2 l/kg for clavulanic acid.

Biotransformation

Amoxicillin is partly excreted in the urine as the inactive penicilloic acid in quantities equivalent to up to 10 to 25% of the initial dose. Clavulanic acid is extensively metabolized in man and eliminated in urine and faeces and as carbon dioxide in expired air.

Elimination

The major route of elimination for amoxicillin is via the kidney, whereas for clavulanic acid it is by both renal and non-renal mechanisms.

Amoxicillin/clavulanic acid has a mean elimination half-life of approximately one hour and a mean total clearance of approximately 25 l/h in healthy subjects. Approximately 60 to 70% of the amoxicillin and approximately 40 to 65% of the clavulanic acid are excreted unchanged in urine during the first 6 h after administration of single Remedica Co-Amoxiclav 250 mg/125 mg or 500 mg/125 mg tablets.

Age

The elimination half-life of amoxicillin is similar for children aged around 3 months to 2 years and older children and adults. For very young children (including preterm newborns) in the first week of life the interval of administration should not exceed twice daily administration due to immaturity of the renal pathway of elimination. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function.

Gender

Following oral administration of amoxicillin/clavulanic acid to healthy males and female subjects, gender has no significant impact on the pharmacokinetics of either amoxicillin or clavulanic acid.

Renal impairment

The total serum clearance of amoxicillin/clavulanic acid decreases proportionately with decreasing renal function. The reduction in drug clearance is more pronounced for amoxicillin than for clavulanic acid, as a higher proportion of amoxicillin is excreted via the renal route. Doses in renal impairment must therefore prevent undue accumulation of amoxicillin while maintaining adequate levels of clavulanic acid (see section Posology and method of administration).

Hepatic impairment

Hepatically impaired patients should be dosed with caution and hepatic function monitored at regular intervals.

5. Clinical particulars

Indications

Remedica Co-Amoxiclav is an antibiotic agent with a notably broad spectrum of activity against the commonly occurring bacterial pathogens in general practice and hospital. The beta-lactamase inhibitory action of clavulanate extends the spectrum of amoxicillin to embrace a wider range of organisms, including many resistant to other beta-lactam antibiotics.

Remedica Co-Amoxiclav should be used in accordance with local official antibiotic-prescribing guidelines and local susceptibility data.

Remedica Co-Amoxiclav oral presentations for twice daily dosing, are indicated for short-term treatment of bacterial infections at the following sites:

- *Upper respiratory tract infections (including ENT)* e.g. tonsillitis, sinusitis, otitis media.
- *Lower respiratory tract infections* e.g. acute exacerbation of chronic bronchitis, lobar and bronchopneumonia.
- *Genito-urinary tract infections* e.g. cystitis, urethritis, pyelonephritis.
- *Skin and soft tissue infections*, e.g. boils, abscesses, cellulitis, wound infections.
- *Bone and joint infections* e.g. osteomyelitis.
- *Dental infections* e.g. dentoalveolar abscess
- *Other infections* e.g. septic abortion, puerperal sepsis, intra-abdominal sepsis.

Susceptibility to Remedica Co-Amoxiclav will vary with geography and time (see *Pharmacological Properties, Pharmacodynamics* for further information). Local susceptibility data should be consulted where available, and microbiological sampling and susceptibility testing performed where necessary.

Posology and method of administration

Posology

Dosage depends on the age and renal function of the patient and the severity of the infection.

Treatment should not be extended beyond 14 days without review.

Remedica Co-Amoxiclav tablets are not recommended in children of 12 years and under.

Adults and children over 12 years

The usual recommended daily dose is:

Mild - Moderate infections	One Remedica Co-Amoxiclav 625 mg tablet every 12 hours.
Severe infections	One Remedica Co-Amoxiclav 625 mg tablet every 8 hours.

Renal impairment

No adjustment in dose is required in patients with creatinine clearance (CrCl) greater than 30 ml/min.

CrCl: 10-30 ml/min	One Remedica Co-Amoxiclav 625 mg tablet every 12 hours
CrCl < 10 ml/min	One Remedica Co-Amoxiclav 625 mg tablet every 24 hours
Haemodialysis	One Remedica Co-Amoxiclav 625 mg tablet every 24 hours, plus a further one tablet during dialysis, to be repeated at the end of dialysis (as serum concentrations of both amoxicillin and clavulanic acid are decreased)

Hepatic impairment

Dose with caution; monitor hepatic function at regular intervals.

Method of Administration

Remedica Co-Amoxiclav is for oral use.

To minimise potential gastrointestinal intolerance, administer at the start of a meal. The absorption of Remedica Co-Amoxiclav is optimized when taken at the start of a meal.

Therapy can be started parenterally and continued with an oral preparation.

Tablets should be swallowed whole without chewing.

Contraindications

Hypersensitivity to the active substances, to any of the penicillins or to any of the excipients.

History of a severe immediate hypersensitivity reaction (e.g. anaphylaxis) to another beta-lactam agent (e.g. a cephalosporin, carbapenem or monobactam).

History of jaundice/hepatic impairment due to amoxicillin/clavulanic acid (see section Side effects).

Warnings and Precautions

Before initiating therapy with amoxicillin and clavulanate potassium, careful enquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, or other beta-lactam agents.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients on penicillin therapy. These reactions are more likely to occur in individuals with a history of penicillin hypersensitivity and in atopic individuals. If an allergic reaction occurs, amoxicillin/clavulanic acid therapy must be discontinued and appropriate alternative therapy instituted.

In the case that an infection is proven to be due to an amoxicillin-susceptible organisms(s) then consideration should be given to switching from amoxicillin/clavulanic acid to amoxicillin in accordance with official guidance.

This presentation of Remedica Co-Amoxiclav is not suitable for use when there is a high risk that the presumptive pathogens have reduced susceptibility or resistance to beta-lactam agents that is not mediated by beta-lactamases susceptible to inhibition by clavulanic acid (e.g. penicillin-insusceptible *S. pneumoniae*).

Convulsions may occur in patients with impaired renal function or in those receiving high doses (see section Side effects).

Amoxicillin/clavulanic acid should be avoided if infectious mononucleosis is suspected since the occurrence of a morbilliform rash has been associated with this condition following the use of amoxicillin.

Concomitant use of allopurinol during treatment with amoxicillin can increase the likelihood of allergic skin reactions.

Prolonged use may occasionally result in overgrowth of non-susceptible organisms.

The occurrence at the treatment initiation of a feverish generalised erythema associated with pustula may be a symptom of acute generalised exanthematous pustulosis (AGEP) (see Section Side effects). This reaction requires Remedica Co-Amoxiclav discontinuation and contra-indicates any subsequent administration of amoxicillin.

Amoxicillin/clavulanic acid should be used with caution in patients with evidence of hepatic impairment (see section Posology and method of administration).

Hepatic events have been reported predominantly in males and elderly patients and may be associated with prolonged treatment. These events have been very rarely reported in children. In all populations, signs and symptoms usually occur during or shortly after treatment but in some cases may not become apparent until several weeks after treatment has ceased. These are usually reversible. Hepatic events may be severe and, in extremely rare circumstances, deaths have been reported. These have almost always occurred in patients with serious underlying disease or taking concomitant medications known to have the potential for hepatic effects (see section Side effects).

Antibiotic-associated colitis has been reported with nearly all antibacterial agents and may range in severity from mild to life threatening (see section Side effects). Therefore, it is important to consider this diagnosis in patients who present with diarrhoea during or subsequent to the administration of any antibiotics. Should antibiotic-associated colitis occur, amoxicillin/clavulanic acid should immediately be discontinued, a physician be consulted and an appropriate therapy initiated. Anti-peristaltic medicinal products are contra-indicated in this situation.

Periodic assessment of organ system functions, including renal, hepatic and haematopoietic function is advisable during prolonged therapy.

Prolongation of prothrombin time has been reported rarely in patients receiving amoxicillin/clavulanic acid. Appropriate monitoring should be undertaken when anticoagulants are prescribed concomitantly. Adjustments in the dose of oral anticoagulants may be necessary to maintain the desired level of anticoagulation (see section Interaction with other medicaments and Side effects).

In patients with renal impairment, the dose should be adjusted according to the degree of impairment (see section Posology and method of administration).

In patients with reduced urine output, crystalluria has been observed very rarely, predominantly with parenteral therapy. During the administration of high doses of amoxicillin, it is advisable to maintain adequate fluid intake and urinary output in order to reduce the possibility of amoxicillin crystalluria. In patients with bladder catheters, a regular check of patency should be maintained (see section “Symptoms and Treatment of Overdose”).

During treatment with amoxicillin, enzymatic glucose oxidase methods should be used whenever testing for the presence of glucose in urine because false positive results may occur with non-enzymatic methods.

The presence of clavulanic acid in Remedica Co-Amoxiclav may cause a non-specific binding of IgG and albumin by red cell membranes leading to a false positive Coombs test.

There have been reports of positive test results using the Bio-Rad Laboratories Platelia *Aspergillus* EIA test in patients receiving amoxicillin/clavulanic acid who were subsequently found to be free of *Aspergillus* infection. Cross-reactions with non-*Aspergillus* polysaccharides and polyfuranoses with Bio-Rad Laboratories Platelia *Aspergillus* EIA test have been reported. Therefore, positive test results in patients

receiving amoxicillin/clavulanic acid should be interpreted cautiously and confirmed by other diagnostic methods.

Interaction with other medicaments

Oral anticoagulants

Oral anticoagulants and penicillin antibiotics have been widely used in practice without reports of interaction. However, in the literature there are cases of increased international normalised ratio in patients maintained on acenocoumarol or warfarin and prescribed a course of amoxicillin. If co-administration is necessary, the prothrombin time or international normalised ratio should be carefully monitored with the addition or withdrawal of amoxicillin. Moreover, adjustments in the dose of oral anticoagulants may be necessary (see section Warning and Precautions and Side effects).

Methotrexate

Penicillins may reduce the excretion of methotrexate causing a potential increase in toxicity.

Probenecid

Concomitant use of probenecid is not recommended. Probenecid decreases the renal tubular secretion of amoxicillin. Concomitant use of probenecid may result in increased and prolonged blood levels of amoxicillin but not of clavulanic acid.

Mycophenolate mofetil

In patients receiving mycophenolate mofetil, reduction in pre-dose concentration of the active metabolite mycophenolic acid (MPA) of approximately 50% has been reported following commencement of oral amoxicillin plus clavulanic acid. The change in pre-dose level may not accurately represent changes in overall MPA exposure.

Pregnancy and Lactation

Pregnancy

Animal studies do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/foetal development, parturition or postnatal development. Limited data on the use of amoxicillin/clavulanic acid during pregnancy in humans do not indicate an increased risk of congenital malformations. In a single study in women with preterm, premature rupture of the foetal membrane it was reported that prophylactic treatment with amoxicillin/clavulanic acid may be associated with an increased risk of necrotising enterocolitis in neonates. Use should be avoided during pregnancy, unless considered essential by the physician.

Breast-feeding

Both substances are excreted into breast milk (nothing is known of the effects of clavulanic acid on the breast-fed infant). Consequently, diarrhoea and fungus infection of the mucous membranes are possible in the breast-fed infant, so that breast-feeding might have to be discontinued. The possibility of sensitisation should be taken into account. Amoxicillin/clavulanic acid should only be used during breast-feeding after benefit/risk assessment by the physician in charge.

Side effects

The most commonly reported adverse drug reactions (ADRs) are diarrhoea, nausea and vomiting.

The ADRs derived from clinical studies and post-marketing surveillance with amoxicillin/clavulanic acid, sorted by MedDRA System Organ Class are listed below.

Infections and infestations	
Mucocutaneous candidosis	Common
Overgrowth of non-susceptible organisms	Not known
Blood and lymphatic system disorders	
Reversible leucopenia (including neutropenia)	Rare
Thrombocytopenia	Rare
Reversible agranulocytosis	Not known
Haemolytic anaemia	Not known
Prolongation of bleeding time and prothrombin time ¹	Not known
Immune system disorders ¹⁰	
Angioneurotic oedema	Not known
Anaphylaxis	Not known
Serum sickness-like syndrome	Not known
Hypersensitivity vasculitis	Not known
Nervous system disorders	
Dizziness	Uncommon
Headache	Uncommon
Reversible hyperactivity	Not known
Convulsions ²	Not known
Aseptic meningitis	Not known
Gastrointestinal disorders	
Diarrhoea	Very common
Nausea ³	Common
Vomiting	Common
Indigestion	Uncommon
Antibiotic-associated colitis ⁴	Not known
Black hairy tongue	Not known
Hepatobiliary disorders	
Rises in AST and/or ALT ⁵	Uncommon
Hepatitis ⁶	Not known
Cholestatic jaundice ⁶	Not known

Skin and subcutaneous tissue disorders ⁷	
Skin rash	Uncommon
Pruritus	Uncommon
Urticaria	Uncommon
Erythema multiforme	Rare
Stevens-Johnson syndrome	Not known
Toxic epidermal necrolysis	Not known
Bullous exfoliative-dermatitis	Not known
Acute generalised exanthemous pustulosis (AGEP) ⁹	Not known
Drug reaction with eosinophilia and systemic symptoms (DRESS)	Very rare
Renal and urinary disorders	
Interstitial nephritis	Not known
Crystalluria ⁸	Not known
¹ See section Warnings and Precautions. ² See section Warnings and Precautions. ³ Nausea is more often associated with higher oral doses. If gastrointestinal reactions are evident, they may be reduced by taking amoxicillin/clavulanic acid at the start of a meal. ⁴ Including pseudomembranous colitis and haemorrhagic colitis (see section Warnings and Precautions). ⁵ A moderate rise in AST and/or ALT has been noted in patients treated with beta-lactam class antibiotics, but the significance of these findings is unknown. ⁶ These events have been noted with other penicillins and cephalosporins (see section Warnings and Precautions). ⁷ If any hypersensitivity dermatitis reaction occurs, treatment should be discontinued (see section Warnings and Precautions). ⁸ See section Symptoms and Treatment of Overdose. ⁹ See section Warnings and Precautions. ¹⁰ See sections Contraindications and Warnings and Precautions.	

Symptoms and Treatment of Overdose

Symptoms and signs of overdose

Gastrointestinal symptoms and disturbance of the fluid and electrolyte balances may be evident. Amoxicillin crystalluria, in some cases leading to renal failure, has been observed (see section Warnings and Precautions).

Convulsions may occur in patients with impaired renal function or in those receiving high doses.

Amoxicillin has been reported to precipitate in bladder catheters, predominantly after intravenous administration of large doses. A regular check of patency should be maintained (see section Warnings and Precautions).

Treatment of intoxication

Gastrointestinal symptoms may be treated symptomatically, with attention to the water/electrolyte balance. Amoxicillin/clavulanic acid can be removed from the circulation by haemodialysis.

6. Pharmaceutical particulars

Storage conditions

Store below 30°C. Protect from light and moisture.

Shelf-life

2 years.

Pack-size available

Aluminium/Aluminium blisters. Pack size of 20 film-coated tablets.

7. Manufacturer

Remedica Ltd.

Aharnon Street, Limassol Industrial Estate, 3056 Limassol, Cyprus

8. Date of revision of the text

03/2023

For internal use only: my-pi-remedica-co-amoxiclav-625mg-fc-tabs-a3